

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
Second Semester 2022-2023
Comprehensive Exam (EC-3 Makeup)

Course No. : AIMLCZG512

Course Title: Deep Reinforcement Learning

Nature of Exam : Open Book

Weightage : 40%

No. of Pages = 2 ;

No. Of Questions = 6;

Duration : --- Hours;

Date of Exam: 06-10-2024 (EN)

Note to Students:

1. Answer all the questions.
 2. Write your name and sign at the end of all the pages.
 3. Assumptions made if any, should be stated clearly at the beginning of your answer.
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Q1) Answer the questions in the same order, in A4 papers , scan & upload

- a. You are faced with three slot machines (arms) in a casino. Machine A has a win rate of 0.3, Machine B has a win rate of 0.5, and Machine C has a win rate of 0.2. If you play each machine 10 times, what is the expected number of wins for each machine, and how should you approach the exploration vs. exploitation dilemma after these initial plays? [2 Marks]
- b. In an epsilon-greedy algorithm with $\epsilon=0.1$, you have the following estimated rewards for three arms after 20 plays: Arm 1: 0.4, Arm 2: 0.5, Arm 3: 0.3. How many times would you expect to explore (choose a random arm) versus exploit (choose the best arm) over 100 plays? [2 Marks]
- c. Using the UCB algorithm, you have played three arms for the following number of times with the respective average rewards: Arm 1 (5 plays, avg reward 0.2), Arm 2 (10 plays, avg reward 0.5), Arm 3 (15 plays, avg reward 0.4). Calculate the UCB for each arm if the total number of plays is 30. Assume a confidence level of $\sqrt{\frac{2\ln(30)}{n}}$ [2 Marks]

[2+2+2 = 6 Marks]

Q2) Answer the questions in the same order, in A4 papers , scan & upload

Imagine you are in charge of an online store where you are experimenting with different promotions (e.g., discounts, free shipping, bonus items) to maximize customer purchases. You can think of this scenario as a multi-armed bandit problem where each "arm" represents a different promotion strategy.

- a. How would you use an Upper Confidence Bound (UCB) algorithm to decide which promotion to show to customers? Explain how UCB balances exploration and exploitation in this context. [3 Marks]
- b. Consider a situation where one promotion shows very high rewards initially, but over time, its performance decreases. What potential problem could arise if you rely on an ϵ -greedy algorithm in this situation, and how might an adaptive strategy like UCB help address this issue? [2 Marks]
- c. How might incorporating customer feedback or demographic data improve the effectiveness of your multi-armed bandit algorithm in selecting the best promotion for different customer segments? [2 Marks]

[3+3+2 = 7 Marks]

Q3) Answer the questions in the same order, in A4 papers , scan & upload

- (a) You are using the Monte Carlo method to estimate the expected return of a reinforcement learning agent that performs in an environment over 10 episodes. The observed returns for each episode for a given state s are as follows: [12,15,20,10,25,18,30,8,22,14]. Given that the returns are subject to a discount factor $\gamma=0.95$, what would be the estimated return for the state s after all the episodes? Explain. [2 Marks] Assume the episodes were generated using some policy. What framework do you use to improve the policy that generated the episode? Explain. [2 Marks]
- (b) Consider a reinforcement learning scenario where an agent interacts with an environment for 4 steps, and the observed rewards for these steps are: [5,10,-2,8]. (i) Compute the cumulative returns for each time step (returns from that step until the end). [1 Marks] (ii) Apply a discount factor $\gamma=0.9$ to compute the discounted returns for each time step. [1 Marks]. (iii) If the agent continues to receive rewards as per the average of the last 4 rewards, estimate the expected return after 10 steps using Monte Carlo simulation. [1 Marks]

[4+3 = 7 Marks]

Q4) Answer the questions in the same order, in A4 papers , scan & upload

- a. Consider a Gridworld shown below where we want to use Temporal Difference (TD0) Learning to estimate the values of the states. Suppose we observe the following state-action transitions and rewards: (S0, North, S3, 2), (S3, East, S4, 4), (S4, East, S5, -10), (S5, West, S4, 8) (S4, North, S7, 6), (S7, East, S8, 10). Assume the initial values of each state are: $V(S0)=V(S1)=V(S2)=V(S3)=V(S4)=V(S5)=V(S6)=V(S7)=V(S8)=0$. The discount factor $\gamma=0.9$ and the learning rate $\alpha=0.5$.

S6	S7	S8
S3	S4	S5 
S0 	S1	S2

Using the TD(0) update rule, what are the learned values for each state after applying TD(0) learning for all six observations? [3 Marks]. Why might Temporal Difference (TD) learning be preferred over Monte Carlo methods in certain situations? Provide at least a reason with an appropriate example. [2 Mark]

- b. Imagine you are programming a robot to navigate a warehouse and pick up items from various locations. The robot must learn to avoid obstacles, plan efficient routes, and adapt to changes in the warehouse layout, such as new shelves or boxes being moved around. You decide to use reinforcement learning to help the robot learn an optimal policy. What difficulties would you encounter if you tried to use tabular learning to represent every possible state the robot might encounter in the warehouse environment? [1 Mark] How could using a parameterized function approximator, where the robot learns high-level tasks like "move to shelf" and "pick up item" separately, help address these challenges in complex environments like the warehouse? [1 Mark]

[5+2 = 7 Marks]

Q5) Answer the questions in the same order, in A4 papers , scan & upload

- a. You are developing an intelligent agent for a strategy board game where the agent must plan multiple steps, adapt to the actions of opponents, and optimize its strategies to win. The state and action spaces are large, and the game dynamics can change depending on the players' strategies.
- In this scenario, how would using the REINFORCE algorithm help the agent learn an optimal policy for making strategic decisions? Identify one potential drawback of using REINFORCE in such a complex, multi-step planning environment. [1 Mark]
 - How could implementing an advantage actor-critic (A2C) method improve the agent's learning performance compared to using REINFORCE? Explain the roles of the actor and the critic in A2C, and how the advantage function reduces variance in policy updates. State how this approach could enhance decision-making in the strategy board game. [1 Mark]
 - Discuss the impact of using a baseline (such as the value function) on the learning stability and convergence speed of the REINFORCE and actor-critic methods. Why might the actor-critic method lead to more stable and efficient learning in the context of this board game? [1 Mark]
- b. Policy gradient methods aim to optimize a policy directly by following the gradient of expected rewards. In environments with **non-stationary dynamics** (i.e., environments where the rules change over time), what potential challenges might a policy gradient method face? How could an algorithm be adapted to handle such non-stationary environments? [2 Marks]
- c. In policy gradient methods, the policy is typically stochastic, meaning it selects actions based on probabilities. How does this stochastic nature benefit exploration compared to deterministic policies? In what scenarios could stochastic policies be particularly advantageous? [2 Marks]

[3+3+2= 7 Marks]

Q6) Answer the questions in the same order, in A4 papers , scan & upload

- a. MCTS relies on simulations to explore possible future game states. How does MCTS balance exploration of less-known moves versus exploitation of moves that have been proven to yield higher rewards? What role does the Upper Confidence Bound (UCB) play in this balance, and why is it important in games with large decision spaces like Go? [1.5 Marks]
- b. Unlike its predecessors, AlphaGo Zero learned to play Go solely through **self-play**, without using human expert games as training data. What advantage does self-play offer compared to using imitation learning from human games? How does this contribute to discovering strategies that even human players might not have thought of? [1.5 Marks]
- c. Consider a node y is selected for expansion and was subsequently evaluated by MCTS. The outcome of the rollout from y results in the return of +1 (i.e. success). How does AlphaGo make an estimate of $v(y)$? [1 Marks] Assume the necessary value of parameters and other estimations and state them clearly. [1 Marks] Explain the role of each of the other values involved in this estimation. [1 Marks].

[1.5+ 1.5+ 3 = 6 Marks]